

Lab: Paper Computer

Goal

To understand how computers follow instructions (Code).

What You Need

- Paper.
- Pencils.
- "Robot" (A parent or friend).

Steps

1. The Code

Write a program for your Robot to draw a Smiley Face.

- "Draw a big circle."
- "Draw two small dots inside at the top."
- "Draw a curve at the bottom."

2. The Execution

Read the code to the Robot. The Robot must do EXACTLY what you say. If you say "Draw a circle" but don't say "Pick up the pencil," the Robot might draw on the table!

3. Debugging

Did the face look right? If not, fix the code. "Pick up pencil. Move to center. Put pencil down."

Discussion

- Computers are not smart. They are obedient.

- You have to be very specific.
- Coding is just giving clear instructions.

What We Learned

Systems only do what we tell them to do.